In [15]:
```
import numpy as np

#Seasons

Seasons = ["2010","2011","2012","2013","2014","2015","2016","2017","2018","2019"]
Sdict = {"2010":0,"2011":1,"2012":2,"2013":3,"2014":4,"2015":5,"2016":6,"2017":7,"2018":8,"20
print(Seasons)
print(Sdict)
```

```
['2010', '2011', '2012', '2013', '2014', '2015', '2016', '2017', '2018', '2019']
{'2010': 0, '2011': 1, '2012': 2, '2013': 3, '2014': 4, '2015': 5, '2016': 6, '2017': 7, '201
8': 8, '2019': 9}
```

In [16]:
```
#Players
Players = ["Sachin","Rahul","Smith","Sami","Pollard","Morris","Samson","Dhoni","Kohli","Sky"]
Pdict = {"Sachin":0,"Rahul":1,"Smith":2,"Sami":3,"Pollard":4,"Morris":5,"Samson":6,"Dhoni":7,
print(Players)
print(Pdict)
```

```
['Sachin', 'Rahul', 'Smith', 'Sami', 'Pollard', 'Morris', 'Samson', 'Dhoni', 'Kohli', 'Sky']
{'Sachin': 0, 'Rahul': 1, 'Smith': 2, 'Sami': 3, 'Pollard': 4, 'Morris': 5, 'Samson': 6, 'Dhon
i': 7, 'Kohli': 8, 'Sky': 9}
```

In [17]:
```
#Salaries
Sachin_Salary = [15946875,17718750,19490625,21262500,23034375,24806250,25244493,27849149,3045
Rahul_Salary = [12000000,12744189,13488377,14232567,14976754,16324500,18038573,19752645,21466
Smith_Salary = [4621800,5828090,13041250,14410581,15779912,14500000,16022500,17545000,1906750
Sami_Salary = [3713640,4694041,13041250,14410581,15779912,17149243,18518574,19450000,22407474
Pollard_Salary = [4493160,4806720,6061274,13758000,15202590,16647180,18091770,19536360,205131
Morris_Salary = [3348000,4235220,12455000,14410581,15779912,14500000,16022500,17545000,190675
Samson_Salary = [3144240,3380160,3615960,4574189,13520500,14940153,16359805,17779458,18668431
Dhoni_Salary = [0,0,4171200,4484040,4796880,6053663,15506632,16669630,17832627,18995624]
Kohli_Salary = [0,0,0,4822800,5184480,5546160,6993708,16402500,17632688,18862875]
Sky_Salary = [3031920,3841443,13041250,14410581,15779912,14200000,15691000,17182000,18673000,
#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Pollard_Salary, Mo
print(Salary)
```

```
[[15946875 17718750 19490625 21262500 23034375 24806250 25244493 27849149
  30453805 23500000]
 [12000000 12744189 13488377 14232567 14976754 16324500 18038573 19752645
  21466718 23180790]
 [ 4621800  5828090 13041250 14410581 15779912 14500000 16022500 17545000
  19067500 20644400]
 [ 3713640  4694041 13041250 14410581 15779912 17149243 18518574 19450000
  22407474 22458000]
 [ 4493160  4806720  6061274 13758000 15202590 16647180 18091770 19536360
  20513178 21436271]
 [ 3348000  4235220 12455000 14410581 15779912 14500000 16022500 17545000
  19067500 20644400]
 [ 3144240  3380160  3615960  4574189 13520500 14940153 16359805 17779458
  18668431 20068563]
 [       0        0  4171200  4484040  4796880  6053663 15506632 16669630
  17832627 18995624]
 [       0        0        0  4822800  5184480  5546160  6993708 16402500
  17632688 18862875]
 [ 3031920  3841443 13041250 14410581 15779912 14200000 15691000 17182000
  18673000 15000000]]
```

In [20]:
```
#Games
Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
```

```python
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G,
print(Games)
```

```
[[80 77 82 82 73 82 58 78  6 35]
 [82 57 82 79 76 72 60 72 79 80]
 [79 78 75 81 76 79 62 76 77 69]
 [80 65 77 66 69 77 55 67 77 40]
 [82 82 82 79 82 78 54 76 71 41]
 [70 69 67 77 70 77 57 74 79 44]
 [78 64 80 78 45 80 60 70 62 82]
 [35 35 80 74 82 78 66 81 81 27]
 [40 40 40 81 78 81 39  0 10 51]
 [75 51 51 79 77 76 49 69 54 62]]
```

In [ ]:

```python
#Points
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samso
```

In [21]:

```python
print(Points)
```

```
[[2832 2430 2323 2201 1970 2078 1616 2133   83  782]
 [1653 1426 1779 1688 1619 1312 1129 1170 1245 1154]
 [2478 2132 2250 2304 2258 2111 1683 2036 2089 1743]
 [2122 1881 1978 1504 1943 1970 1245 1920 2112  966]
 [1292 1443 1695 1624 1503 1784 1113 1296 1297  646]
 [1572 1561 1496 1746 1678 1438 1025 1232 1281  928]
 [1258 1104 1684 1781  841 1268 1189 1186 1185 1564]
 [ 903  903 1624 1871 2472 2161 1850 2280 2593  686]
 [ 597  597  597 1361 1619 2026  852    0  159  904]
 [2040 1397 1254 2386 2045 1941 1082 1463 1028 1331]]
```

In [22]:

```python
Salary/Games # Here output is in float datatype
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_11604\3709746658.py:1: RuntimeWarning: divide by ze
ro encountered in divide
  Salary/Games
```

Out[22]:
```
array([[ 199335.9375    ,  230113.63636364,  237690.54878049,
         259298.7804878 ,  315539.38356164,  302515.24390244,
         435249.87931034,  357040.37179487, 5075634.16666667,
         671428.57142857],
       [ 146341.46341463,  223582.26315789,  164492.40243902,
         180159.07594937,  197062.55263158,  226729.16666667,
         300642.88333333,  274342.29166667,  271730.60759494,
         289759.875     ],
       [  58503.79746835,   74719.1025641 ,  173883.33333333,
         177908.40740741,  207630.42105263,  183544.30379747,
         258427.41935484,  230855.26315789,  247629.87012987,
         299194.20289855],
       [  46420.5       ,   72216.01538462,  169366.88311688,
         218342.13636364,  228694.37681159,  222717.44155844,
         336701.34545455,  290298.50746269,  291006.15584416,
         561450.        ],
       [  54794.63414634,   58618.53658537,   73917.97560976,
         174151.89873418,  185397.43902439,  213425.38461538,
         335032.77777778,  257057.36842105,  288918.        ,
         522835.87804878],
       [  47828.57142857,   61380.        ,  185895.52238806,
         187150.4025974 ,  225427.31428571,  188311.68831169,
         281096.49122807,  237094.59459459,  241360.75949367,
         469190.90909091],
       [  40310.76923077,   52815.        ,   45199.5       ,
          58643.44871795,  300455.55555556,  186751.9125    ,
         272663.41666667,  253992.25714286,  301103.72580645,
         244738.57317073],
       [      0.        ,       0.        ,   52140.        ,
          60595.13513514,   58498.53658537,   77611.06410256,
         234948.96969697,  205797.90123457,  220155.88888889,
         703541.62962963],
       [      0.        ,       0.        ,       0.        ,
          59540.74074074,   66467.69230769,   68471.11111111,
         179325.84615385,             inf, 1763268.8       ,
         369860.29411765],
       [  40425.6       ,   75322.41176471,  255710.78431373,
         182412.41772152,  204933.92207792,  186842.10526316,
         320224.48979592,  249014.49275362,  345796.2962963 ,
         241935.48387097]])
```

In [23]:
```
Salary//Games # Here you get output in integer datatype
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_11604\1634212085.py:1: RuntimeWarning: divide by ze
ro encountered in floor_divide
  Salary//Games
```

Out[23]:
```
array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
         357040, 5075634,  671428],
       [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
         274342,  271730,  289759],
       [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
         230855,  247629,  299194],
       [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
         290298,  291006,  561450],
       [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
         257057,  288918,  522835],
       [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
         237094,  241360,  469190],
       [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
         253992,  301103,  244738],
       [      0,       0,   52140,   60595,   58498,   77611,  234948,
         205797,  220155,  703541],
       [      0,       0,       0,   59540,   66467,   68471,  179325,
              0, 1763268,  369860],
       [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
         249014,  345796,  241935]])
```

In [25]:
```
Games[:5] # Displays 5 rows from the Games table
```

Out[25]:
```
array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
       [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
       [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
       [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
       [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]])
```

In [26]:
```
Salary[0] # This displays the salry of "Sachin".
```

Out[26]:
```
array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
       25244493, 27849149, 30453805, 23500000])
```

In [29]:
```
Pdict
```

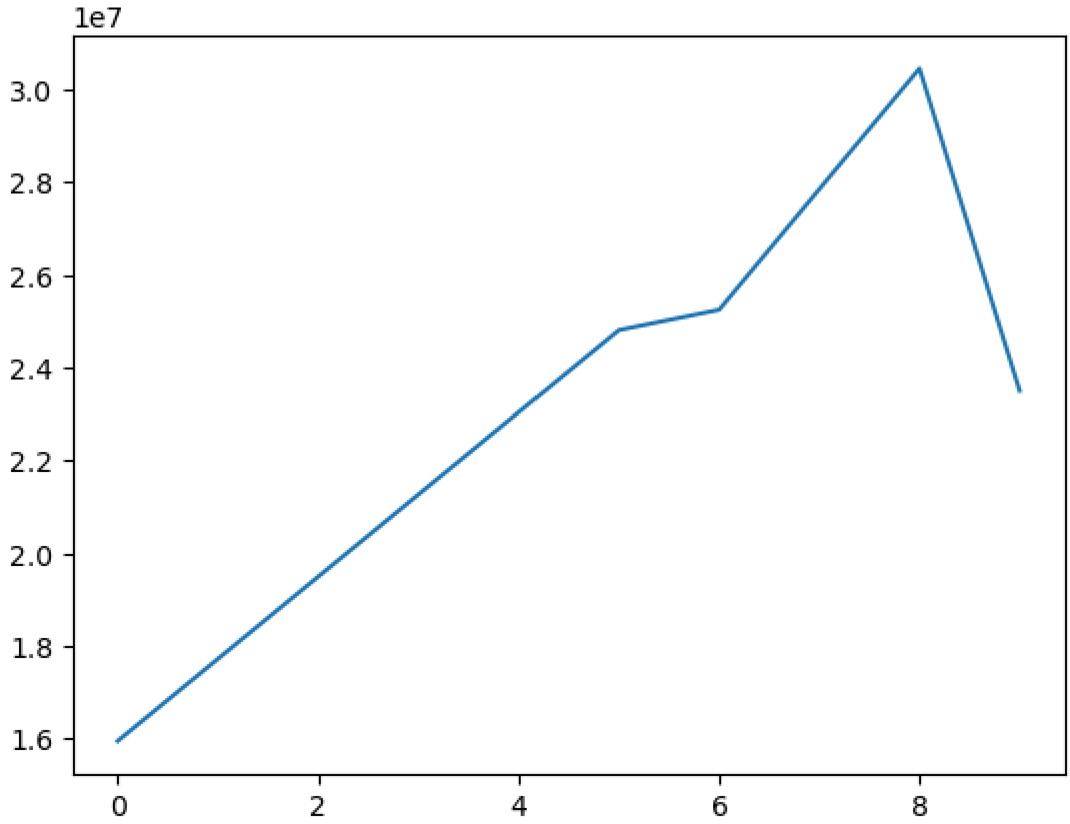
Out[29]:
```
{'Sachin': 0,
 'Rahul': 1,
 'Smith': 2,
 'Sami': 3,
 'Pollard': 4,
 'Morris': 5,
 'Samson': 6,
 'Dhoni': 7,
 'Kohli': 8,
 'Sky': 9}
```

In [30]:
```
import warnings                    # It ignore os warning error while visualize the graph
warnings.filterwarnings('ignore')
```

In [31]:
```
import matplotlib.pyplot as plt
```

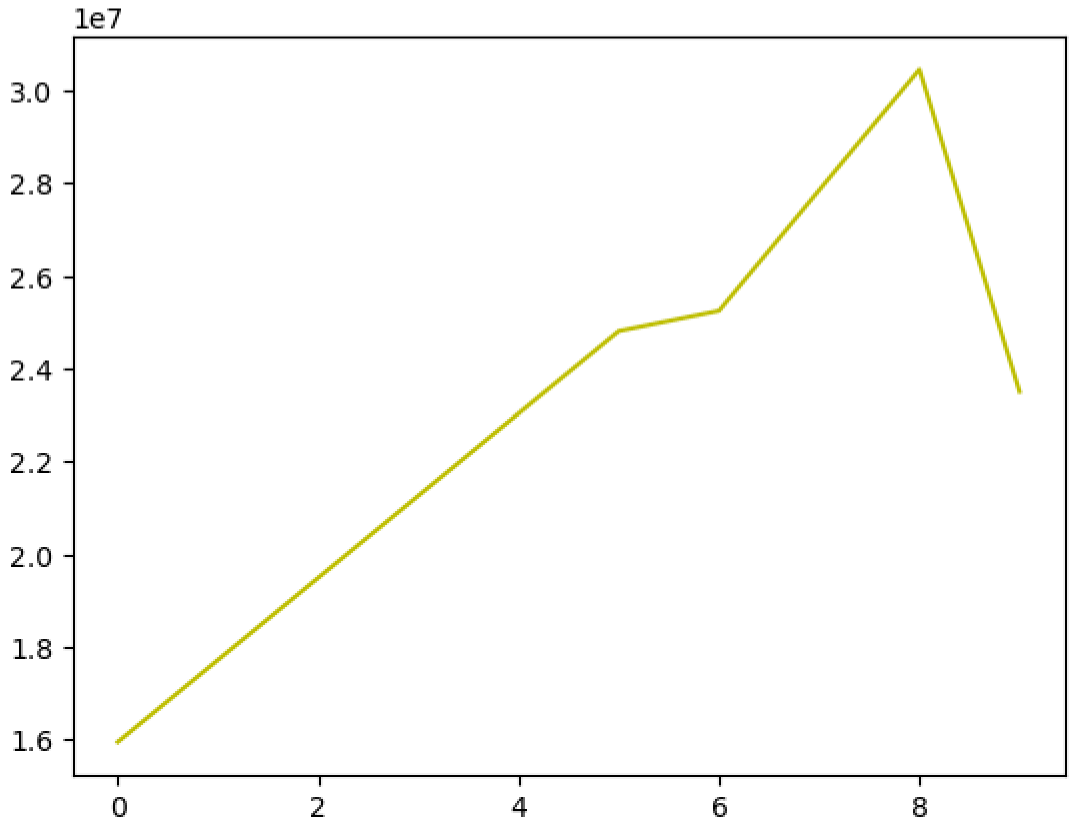
In [45]:
```
plt.plot(Salary[0])
```

Out[45]:
```
[<matplotlib.lines.Line2D at 0x1557b51a350>]
```



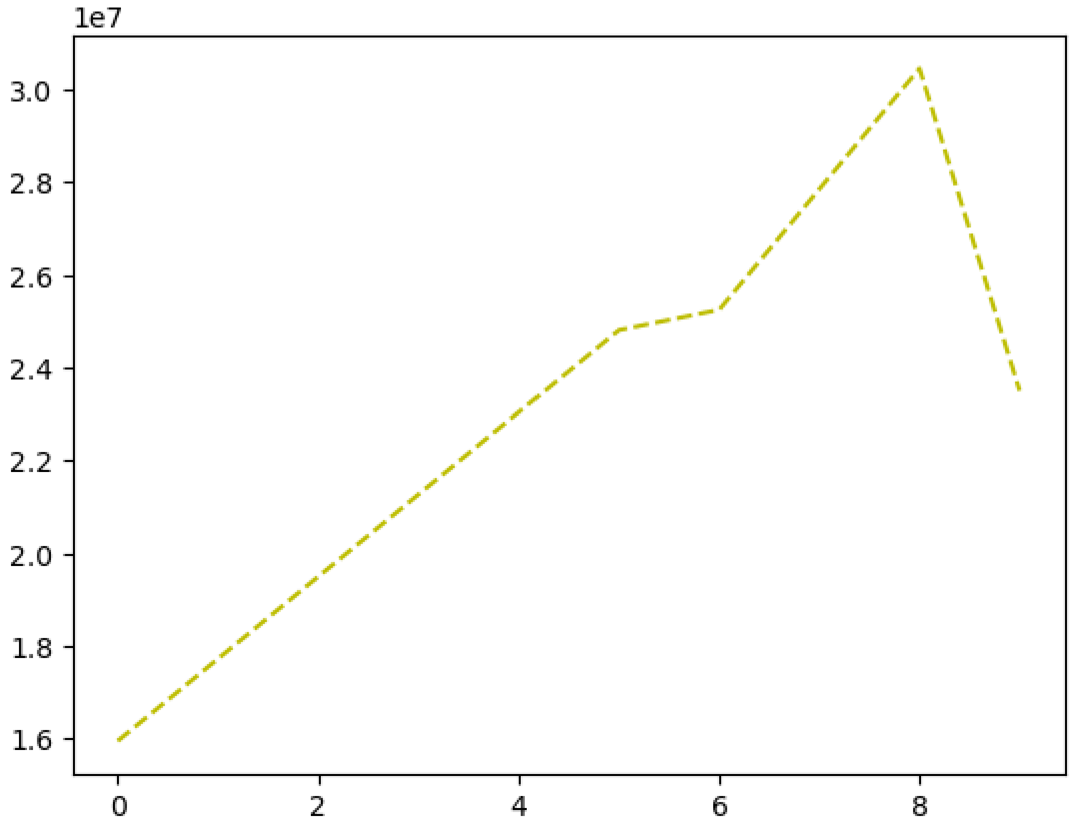
In [46]:
```
plt.plot(Salary[0], c = 'y')
```

Out[46]: [<matplotlib.lines.Line2D at 0x1557b544190>]



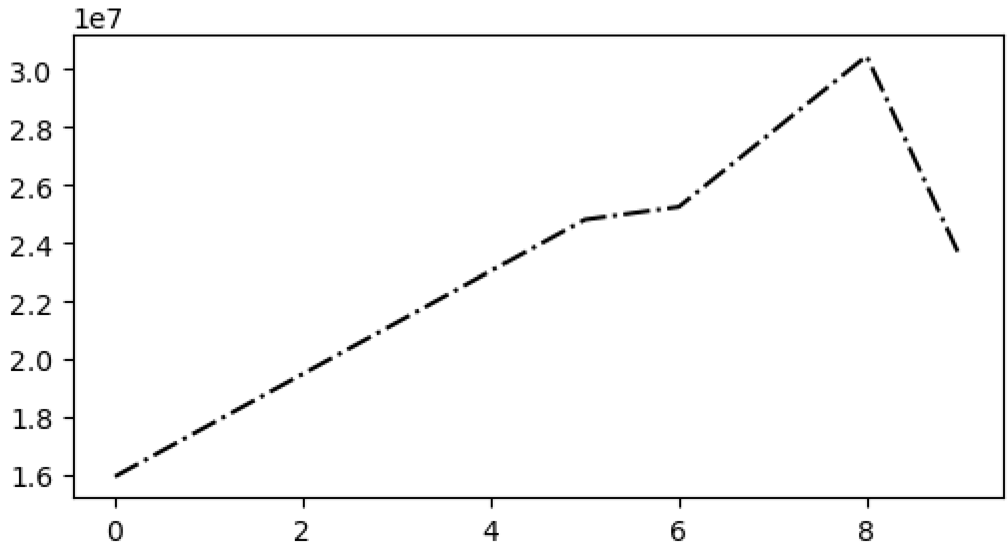
In [48]:
```
plt.plot(Salary[0], c='y', ls = '--')
```

Out[48]: [<matplotlib.lines.Line2D at 0x1557b64ee90>]



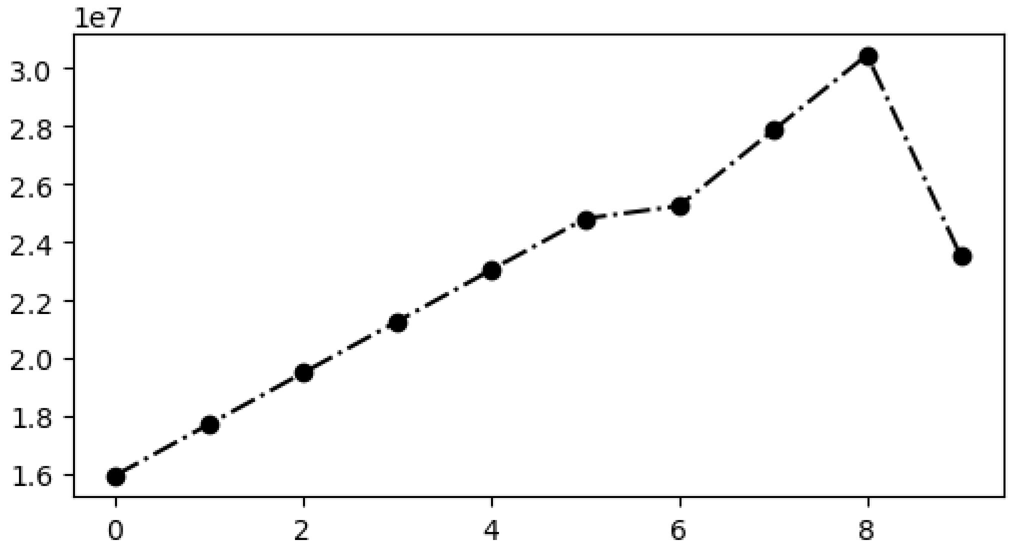
In [63]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = 6,3
```

In [69]:
```
plt.plot(Salary[0], c='k', ls = '-.')
plt.show()
```

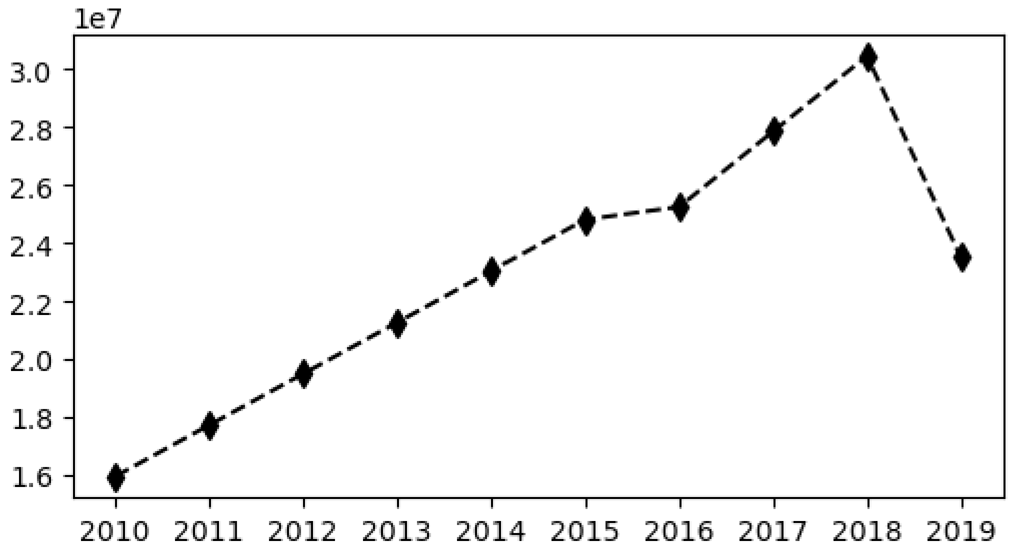


In [71]:
```
plt.plot(Salary[0], c='k', ls = '-.', marker = 'o')
```

Out[71]: `[<matplotlib.lines.Line2D at 0x1557a156350>]`



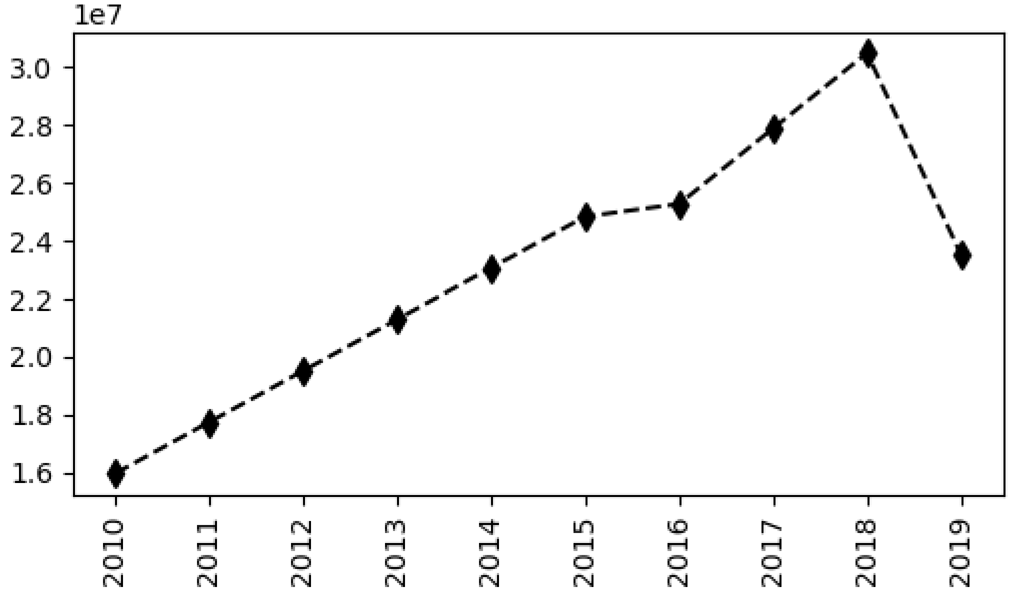
In [83]:
```
plt.plot(Salary[0], c='black', ls = '--', marker = 'd', ms = 7)  # 'd' dimand shape
plt.xticks(list(range(0,10)), Seasons)
plt.show()
```



In [84]:
```
Games[0]
```

Out[84]:
```
array([80, 77, 82, 82, 73, 82, 58, 78,  6, 35])
```

In [93]:
```
plt.plot(Salary[0], c='black', ls = '--', marker = 'd', ms = 7)
plt.xticks(list(range(0,10)), Seasons, rotation='vertical')
plt.show()
```



1e7
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019

In [ ]: